

Literature Review

DocShield AI: An AI-Powered Document and Certificate Forgery Detection System

A Java Spring Boot-based hybrid verification framework combining OCR, computer vision, metadata forensics, and QR-based validation

1. Introduction

Documents and certificates have moved almost entirely into digital form over the last decade — academic degrees, offer letters, government IDs, and professional certifications are now routinely issued, shared, and stored as PDFs or scanned images rather than paper originals. This shift has made verification faster and more convenient, but it has also lowered the barrier to forgery. Widely available image editing and PDF authoring tools make it possible to alter names, dates, grades, or seals with very little skill, and the resulting fake documents are often visually indistinguishable from genuine ones.

Reported cases of forged degree certificates, fake experience letters, and tampered mark-sheets have increased alongside this ease of editing, creating real risk for universities, employers, and certifying bodies who rely on the honesty of the documents presented to them. Manual verification — physically checking a certificate against institutional records — does not scale to this problem. It is slow, requires a human expert who may not recognise a well-made forgery, and is impractical when hundreds or thousands of documents must be checked, for example during bulk recruitment drives or admission cycles.

Advances in Artificial Intelligence (AI) and Optical Character Recognition (OCR) offer a practical alternative. OCR can extract and cross-check the textual content of a document automatically, while AI-based computer vision can detect the subtle visual inconsistencies — mismatched fonts, cloned regions, recompression artefacts — left behind by editing software. Combined with metadata inspection and QR-based cross-verification, these techniques make it possible to build automated systems that flag suspicious documents for human review rather than requiring every document to be checked by hand. This literature review surveys the existing technologies, research, and commercial systems in this space, identifies the gaps that remain, and positions the proposed project, DocShield AI, as a response to those gaps.

2. Review of Existing Technologies

Automated document verification draws on several distinct but complementary technologies. Each addresses a different aspect of forgery — textual, visual, structural, or contextual — and no single one is sufficient on its own. This section reviews the five technology areas most relevant to DocShield AI.

2.1 Optical Character Recognition (OCR)

OCR is the technology that converts the text embedded in an image — a scanned certificate, a photographed mark-sheet — into machine-readable, editable text. Internally, an OCR engine typically segments the image into lines and characters, then uses a trained recognition model (historically pattern-matching, now largely neural network-based) to map each character shape to its corresponding letter or digit. Once text is extracted, it can be searched, compared, or cross-checked against a source-of-truth database.

Tesseract OCR is the most widely used open-source engine in academic and hobbyist projects, and is directly relevant to DocShield AI since it is available as a Java-native wrapper (Tess4J). Since version 4, Tesseract has used an LSTM (Long Short-Term Memory) neural network for line recognition while retaining its legacy pattern-based engine for compatibility.

Advantages: OCR is fast, free, well-documented, and works offline, which matters for a project that must avoid recurring cloud costs. It is highly effective on clean, high-resolution scans of printed text.

Limitations: accuracy drops sharply on low-quality scans, skewed or rotated images, unusual fonts, or handwritten sections. OCR on its own also cannot detect whether the underlying image has been visually tampered with — it only reads whatever text is present, forged or not — which is precisely why OCR must be paired with computer vision and metadata analysis rather than used alone.

2.2 Computer Vision for Tampering Detection

Where OCR reads text, computer vision inspects the image itself for signs of manipulation. Image processing techniques such as noise analysis, Error Level Analysis (ELA), and edge/texture inconsistency detection can reveal that a region of an image was copied, pasted, or re-saved at a different compression level than the rest of the document — a common artefact of certificate editing.

Fake image and copy-move forgery detection has become a major research area, with deep learning approaches (particularly Convolutional Neural Networks) now outperforming older block-matching and keypoint-based methods on benchmark datasets such as CASIA and MICC-F220. These models learn to recognise the subtle statistical fingerprints that manual editing leaves behind, even when the result looks convincing to the human eye.

OpenCV is the standard open-source library for implementing these techniques, and its Java bindings make it a natural fit for a Spring Boot-based system: it provides ready-made functions for edge detection, contour analysis, and image comparison that can be combined with a custom-trained tampering-classification model (via DJL — Deep Java Library — in DocShield AI's case) without needing a Python microservice.

Advantages: computer vision can catch visual forgeries that OCR and metadata checks miss entirely, such as a photoshopped signature or an altered grade that was re-typeset to match the original font.

Limitations: CNN-based detectors need reasonably large, labelled training datasets to generalise well, and can be computationally heavier than OCR or metadata checks, which is a real constraint for a lightweight, self-hosted college project.

2.3 Metadata Analysis

Every digital file carries metadata — descriptive information stored alongside the visible content. For images this is typically EXIF (Exchangeable Image File Format) data: capture device, timestamps, software used, and modification history; for PDFs it includes creation and modification dates, producing application, and author fields.

Metadata matters because editing software leaves traces even when the visual content is altered convincingly. A certificate whose 'creation date' is later than the event it certifies, or whose 'producer' field shows an image editor rather than the issuing institution's document system, is a strong signal of tampering. Research in this area has shown that comparing EXIF fields, thumbnail data, and compression signatures before and after editing is an effective way to flag manipulated files, and that inconsistencies between an image's declared metadata and its actual content are a reliable, low-cost tampering indicator.

Advantages: metadata checks are computationally cheap, require no training data, and can be implemented quickly using existing Java libraries. Limitations: metadata can be stripped or forged by a sufficiently determined actor, and many legitimate transfer channels (chat apps, some cloud uploads) also strip metadata, so it must be treated as one signal among several rather than definitive proof.

2.4 QR Code Verification

A growing number of certificate-issuing systems embed a QR code on the document itself, encoding a certificate ID, a hash of the certificate's contents, or a verification URL. A verifier scans the code and the system checks the encoded data against a trusted record — often stored on a blockchain or in an institutional database — to confirm the certificate was genuinely issued and has not been altered since.

This approach has been explored extensively in recent academic literature on blockchain-based degree verification, where a QR code linked to a blockchain transaction or smart-contract hash allows instant, tamper-evident validation without contacting the issuing institution directly.

Advantages: QR verification is fast, intuitive for end users, and provides strong tamper-evidence when the embedded data is cryptographically linked to the original content. Limitations: it only works for documents that were issued with an embedded QR code in the first place — it is useless against older certificates or ones from institutions that do not use the scheme — and does not, by itself, detect visual tampering of the printed content around the QR code.

2.5 AI in Document Verification

Beyond the specific techniques above, Machine Learning (ML) and Deep Learning (DL) are increasingly used to combine multiple weak signals into a single, holistic authenticity judgement. Document classification models can identify the type of document and check whether its layout

matches the expected template for that issuer; fraud-detection models can be trained on a combination of OCR-extracted fields, image-tampering scores, and metadata flags to output a single confidence or authenticity score.

This 'multi-signal fusion' idea — rather than relying on any one technique — is the central design philosophy behind DocShield AI's proposed authenticity score, and is consistent with the direction recent literature has taken as single-technique systems (pure OCR, pure metadata) have proven insufficient against modern forgery methods.

3. Existing Systems

Several commercial and platform-level solutions already address parts of the document verification problem. Reviewing them clarifies what DocShield AI can learn from, and where a gap remains for a free, self-hosted, education-focused system.

3.1 Adobe Acrobat Digital Signature Verification

Adobe Acrobat allows a PDF to be digitally signed using a certificate-based signature, and can automatically verify whether a signed PDF has been altered since signing by checking the cryptographic signature against the document's current hash.

This provides very strong, mathematically verifiable tamper-evidence, but only for documents that were digitally signed at creation using Adobe's or a compatible signing workflow. It does nothing for the vast majority of certificates already in circulation as plain scanned images or unsigned PDFs, and it cannot assess whether a document was fraudulent from the moment it was issued — it only proves whether a signed file has changed afterward.

3.2 Google Document AI

Google Document AI provides excellent OCR and document understanding capabilities, including pre-trained parsers for specific document types such as invoices and identity documents. However, it primarily focuses on information extraction rather than authenticity or forgery detection, and it requires cloud services and paid per-page API calls, making it costly and less suitable for a standalone educational application that needs to run offline or at zero recurring cost.

3.3 Microsoft Azure AI Document Intelligence

Azure AI Document Intelligence (formerly Form Recognizer) offers similar strengths: high-accuracy OCR, layout analysis, and structured field extraction from forms and IDs, with prebuilt models for common document types. Like Google's offering, it is a cloud-hosted, subscription-billed service. It is well suited to enterprise-scale document processing pipelines, but its cost structure and dependency on continuous internet connectivity make it a poor fit for a free student or small-institution project, and it does not natively provide an image-tampering or copy-move forgery score — its confidence scores relate to extraction quality, not authenticity.

3.4 Amazon Textract

Amazon Textract extracts text, forms, and tables from scanned documents with strong accuracy, and integrates well with other AWS services for downstream processing. Its limitations mirror the other cloud offerings: usage-based pricing that scales poorly for a project with no revenue, a hard dependency on AWS infrastructure, and a scope limited to extraction rather than forensic authenticity checks such as tamper detection, metadata analysis, or QR-based cross-verification.

3.5 Comparative Summary

System	Core Strength	Key Limitation
Adobe Acrobat Digital Signatures	Cryptographically verifiable tamper-evidence after signing	Only works for digitally signed PDFs; no forgery detection for scans/images
Google Document AI	High-accuracy OCR and document understanding	Cloud-only, paid API, no authenticity/tampering scoring
Azure AI Document Intelligence	Strong layout and form-field extraction	Subscription cost, requires connectivity, no tamper detection
Amazon Textract	Reliable text/table/form extraction, AWS integration	Usage-based cost, extraction-only scope, no metadata or QR checks

Across all four systems, a consistent pattern emerges: they are excellent at extracting information from documents, but none of them is designed to answer the question DocShield AI is built around — is this specific document authentic? — and none of them combines OCR with image forensics, metadata analysis, and QR verification in a single free, offline-capable tool.

4. Research Gap

The technologies and systems reviewed above show that strong individual pieces of the document-verification problem already exist — accurate OCR, capable tampering-detection models, metadata forensics techniques, and QR/blockchain verification schemes. What is missing, both in commercial systems and in most of the individual research papers reviewed for this project, is their integration into a single, cohesive, low-cost pipeline aimed specifically at certificates and academic/professional documents.

4.1 Limitations of Current Systems

- Depend on cloud services: Google Document AI, Azure AI Document Intelligence, and Amazon Textract all require continuous internet access and vendor lock-in to a specific cloud provider.
- Require paid APIs: per-page or subscription pricing makes these tools expensive for small colleges, student organisations, or individual verifiers who need to check documents in bulk without a budget.
- Focus mainly on OCR and extraction: the commercial systems are optimised for pulling structured data out of documents, not for judging whether the document is genuine.
- Do not integrate metadata analysis: none of the reviewed commercial systems inspects EXIF/PDF metadata as part of its output, despite this being a cheap and effective forgery signal in the research literature.
- Limited image tampering detection: copy-move and splicing detection research (CNN-based models, ELA) exists as standalone academic work but is rarely combined with OCR or metadata in a single deployable tool.
- Expensive for small organisations: the recurring API costs of cloud OCR/document-AI services scale badly for institutions issuing or checking thousands of certificates with no dedicated budget for verification software.

4.2 How DocShield AI Addresses the Gap

DocShield AI is designed specifically to close these gaps by combining, in a single Java-native application, the techniques that existing systems and papers treat separately:

- Uses Java Spring Boot: a single backend framework for the whole pipeline, avoiding the fragmentation of mixing Python microservices with a Java front end.
- Uses free AI libraries: Tess4J (OCR), OpenCV/JavaCV (computer vision), ZXing (QR codes), and DJL (tampering-classification model) — no paid APIs and no mandatory cloud dependency.
- Combines OCR, image analysis, metadata analysis, and QR verification into one authenticity pipeline rather than treating them as separate tools.
- Produces a single Authenticity Score that fuses signals from all four techniques, giving verifiers one interpretable number instead of four disconnected reports.
- Generates a PDF report suitable for institutional record-keeping, something none of the four commercial systems reviewed does natively for authenticity findings.

In short, the existing literature and commercial landscape confirm that each individual technique DocShield AI uses is well established and effective on its own, but also confirm that no accessible, free, Java-based system currently fuses them into one certificate-forgery-detection pipeline — which is exactly the gap this project is positioned to fill.

5. Summary

This review examined five core technology areas relevant to document and certificate verification — OCR, computer vision-based tampering detection, metadata analysis, QR code verification, and AI/ML-based document classification and fraud detection — alongside four leading commercial systems (Adobe Acrobat, Google Document AI, Azure AI Document Intelligence, and Amazon Textract) and a cross-section of recent (2021–2026) research papers on Tesseract OCR, CNN-based image forgery detection, EXIF metadata forensics, and blockchain/QR-based certificate verification.

The consistent finding across both the commercial and academic literature is that individual techniques are mature and well-validated in isolation, but that combining them into a single, affordable, self-hosted verification pipeline — particularly one built natively in Java rather than requiring a separate Python or cloud service — remains largely unaddressed. Existing commercial tools are powerful but costly and extraction-focused; existing research papers are rigorous but typically validate one technique at a time on benchmark datasets rather than building an end-to-end usable system for institutions.

This gap is precisely why DocShield AI is needed: an affordable, Java Spring Boot-based system that fuses OCR, image tampering detection, metadata forensics, and QR verification into a single authenticity score and report, giving small colleges, hackathon organisers, and individual verifiers a practical tool that the current landscape does not otherwise provide.

6. Literature Review Table

The table below summarises the papers reviewed for this project, spanning OCR, image/certificate forgery detection, metadata forensics, and QR-based verification.

#	Paper / Study	Method Used	Advantages	Limitations
1	Smith, "An Overview of the Tesseract OCR Engine", ICDAR 2007 (foundational; still the basis of Tess4J used in DocShield AI)	LSTM-based line recognition + legacy pattern OCR engine	Fast, free, offline, widely supported in Java via Tess4J	Accuracy drops on low-quality, skewed, or handwritten scans
2	Study on OCR using Tesseract (IJRASET, 2022)	Tesseract + LSTM neural OCR for text classification	High accuracy on printed documents; easy Android/Java integration	Struggles with non-standard fonts and noisy backgrounds
3	Hosny et al., efficient CNN for copy-move forgery detection (2022)	Lightweight CNN architecture	Computationally efficient; good accuracy on benchmark datasets	Needs a reasonably large labelled training dataset
4	Diwan et al., SuperPoint keypoint-based copy-move detection (2023)	Keypoint extraction + RANSAC/Fuzzy C-Means outlier removal	Strong localisation of tampered regions; evaluated on CASIA/MICC-F220	Sensitive to post-processing attacks (blurring, recompression)
5	Gangwar & Pathania, EXIF metadata and thumbnail-based tampering detection	EXIF, thumbnail, and compression-signature comparison	Detects edited files cheaply with no model training required	Cannot detect visual tampering when metadata is stripped or forged
6	Detection Techniques of Digital Image Forgery Using Image Metadata (2015; widely cited baseline)	Combined EXIF analysis + Error Level Analysis (ELA)	Improves on metadata-only checks by adding a visual signal	Still limited against forgeries that regenerate consistent metadata
7	Blockchain-based Academic Certificate Verification using QR Code and DApps (2024)	QR code linked to Ethereum blockchain smart contract	Near-instant, tamper-evident verification; no server maintenance cost	Only works for certificates issued through the blockchain scheme; needs institutional adoption
8	Tamper Proof Certificate Verification Using Blockchain (IJCRT, 2024)	SHA-256 hashing + smart contracts + QR/ID lookup portal	Decentralised, immutable record of certificate authenticity	Requires blockchain infrastructure and issuer participation; no OCR/image-level checks

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Adobe Inc. Adobe Acrobat Digital Signatures — Product Documentation.

Google Cloud. Document AI — Product Documentation.

Microsoft Azure. Azure AI Document Intelligence — Product Documentation.

Amazon Web Services. Amazon Textract — Product Documentation.